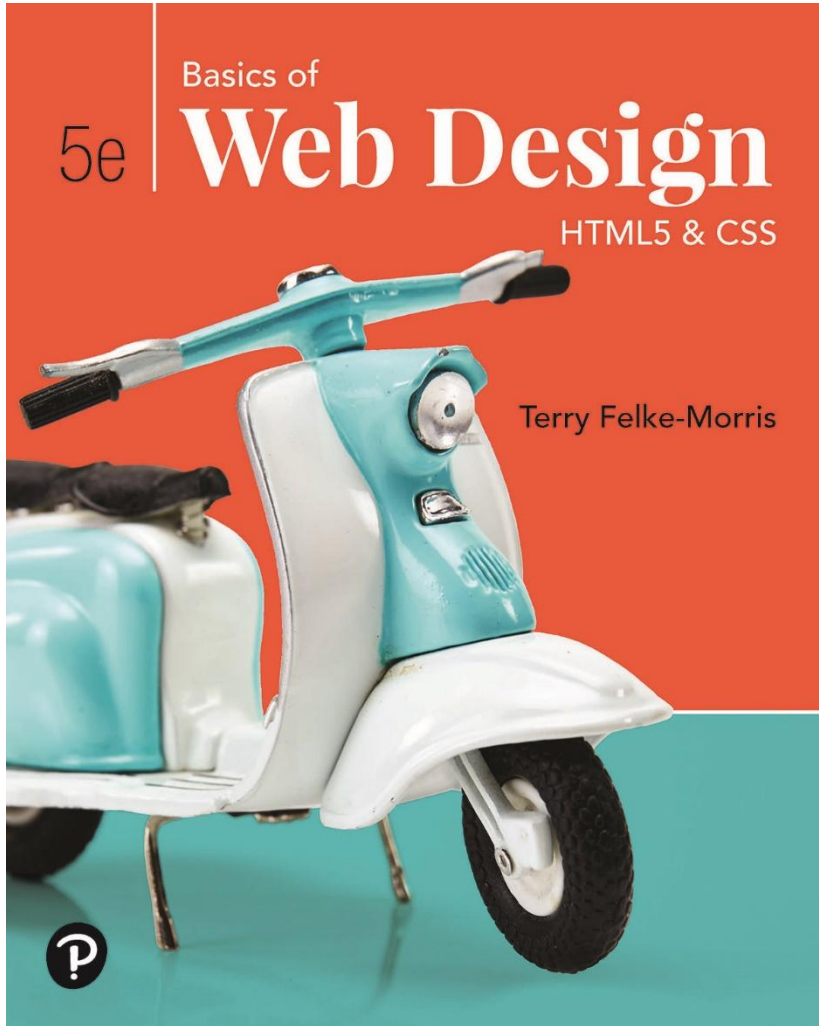


Basics of Web Design

Fifth Edition



Chapter 10

Form Basics Key Concepts

Learning Outcomes

- Describe common uses of forms on web pages
- Create forms on web pages using the form, input, textarea, and select elements
- Associate form controls and groups using label, fieldset, and legend elements
- Use CSS to style a form
- Use Grid Layout to style a form
- Describe the features and common uses of server-side processing
- Invoke server-side processing to handle form data
- Configure new HTML5 form controls and attributes

Overview of Forms

- Forms are used all over the Web to
 - Accept information
 - Provide interactivity
- Types of forms:
 - Search form, Order form, Newsletter sign-up form, Survey form, Add to Cart form, and so on...

Form Definition

HTML form facilitates the user to enter data that is to be sent to the server for processing such as name, email address, password, phone number, etc.

Contains:

controls such as text fields, password fields, checkboxes, radio buttons, submit button, menus etc.

www.Javapoint.com

Two Components of Using Forms

1. The web page form

An HTML element that contains and organizes **form controls** such as text boxes, check boxes, and buttons that can accept information from website visitors.

and

2. The server-side processing

Server-side processing works with the form data and sends e-mail, writes to a text file, updates a database, or performs some other type of processing on the server.

Form Attributes

The **<form>** element supports several attributes that control its behavior:

- **action**: Specifies the URL of the file that will process the form data when submitted.
- If omitted, the data is sent to the current page.
- **target**: Defines where the response from submitting the form will be displayed.
- Common values include `_blank` (new window/tab), `_self` (current window/tab, default), `_parent`, `_top`, or a specific frame name.
- **method**: Determines the HTTP method used to send the form data.

- **get**: Appends form data to the URL as name/value pairs.

This is visible in the URL and has size limitations. **Never use for sensitive data.**

- **post**: Sends form data in the body of the HTTP request.

This method is more secure, has no size limitations, and the data isn't visible in the URL.

- **autocomplete**: Controls whether the browser should automatically complete values based on user's past entries (on or off).
- **novalidate**: A boolean attribute that, when present, prevents the form data from being validated upon submission.

- **name**: Specifies a name for the form.

Form Elements

HTML forms are built using various elements to gather different types of input:

- **<input>**: The most versatile form element, used to create various input fields depending on its type attribute (e.g., text, password, submit, radio buttons, checkboxes).
- **<label>**: Defines a caption for a form element. It improves accessibility by associating text with input fields, especially for screen readers and users who struggle with small clickable areas.
- **<select>**: Creates a drop-down list from which users can choose one or multiple options.
 - **<option>**: Defines an individual option within a `<select>` drop-down list.
 - **<optgroup>**: Groups related options in a drop-down list.
- **<textarea>**: Defines a multi-line text input field, useful for longer text entries like comments or messages.
- **<button>**: Defines a clickable button, which can be used to submit forms, reset forms, or trigger JavaScript functions.
- **<fieldset>**: Used to group related form elements together, visually creating a box around them.
- **<legend>**: Provides a caption or title for the content within a `<fieldset>` element.
- **<datalist>**: Specifies a list of pre-defined options for an `<input>` element, offering suggestions to the user as they type.
- **<output>**: Represents the result of a calculation or user action within a form.

Input Types

HTML forms are essential for collecting user input on websites.

They allow users to interact with the site by entering text, selecting options, uploading files, and more. The data collected is then typically sent to a server for processing.

The `<input>` element's `type` attribute determines the kind of input control it displays:

- text**: A single-line text input field.
- password**: A single-line input field where characters are masked (shown as asterisks or circles).
- submit**: A button for submitting the form data.
- reset**: A button that resets all form values to their default.
- radio**: Allows a user to select only one option from a limited set of choices.
- checkbox**: Allows a user to select zero or more options from a limited set of choices.
- button**: A clickable button that can trigger JavaScript.
- color**: A field for inputting a color, often displaying a color picker.
- date**: A field for selecting a date, often displaying a date picker.
- datetime-local**: A field for selecting a date and time (without time zone).
- email**: A field for entering an email address, with built-in validation for email format.
- file**: A field and "Browse" button for uploading files.
- hidden**: A non-visible input field used to store data that needs to be submitted with the form but not seen or modified by the user. **Not for security.**
- image**: An image that functions as a submit button.

Input Types continue...

- **month**: A field for selecting a month and year.
- **number**: A field for entering a numerical value.
- **range**: A slider control for entering a number within a specified range where the exact value is not known.
- **search**: A text field specifically designed for search queries.
- **tel**: A field for entering a telephone number.
- **time**: A field for selecting a time (without time zone).
- **url**: A field for entering a URL, with built-in validation for URL format.
- **week**: A field for selecting a week and year.

Input Types Attributes

Beyond the type attribute, `<input>` elements can have additional attributes to control their behavior and add restrictions:

- checked**: Specifies that a checkbox or radio button should be pre-selected when the page loads.
- disabled**: Makes an input field unusable and unclickable.
- max**: Specifies the maximum value allowed for numeric or date inputs.
- maxlength**: Specifies the maximum number of characters allowed for text input.
- min**: Specifies the minimum value allowed for numeric or date inputs.
- readonly**: Makes an input field uneditable by the user.
- required**: Specifies that the input field must be filled out before submitting the form.
- size**: Specifies the visible width of the input field in characters.
- step**: Specifies the legal number intervals for numeric or range inputs.
- value**: Specifies the default value for an input field.
- name**: **Crucially important**, this attribute provides the name for the input field, which is used to identify the data when it's submitted to the server. If omitted, the input field's value won't be sent.
- id**: A unique identifier for the input element, often used with the `<label>` element's `for` attribute to associate them.

Form-Specific Attributes

These attributes override the corresponding attributes of the parent `<form>` element for a specific input:

- form**: Specifies the ID of the `<form>` element the `<input>` belongs to,

especially useful if the input is outside the `<form>` tags.

- formaction**: Specifies the URL of the file that will process the input when the form is submitted,

overriding the form's action attribute. Works with submit and image types.

overriding the form's enctype attribute. Works with submit and image types.

- formmethod**: Defines the HTTP method for sending form data, overriding the form's method attribute.

Works with submit and image types.

- formtarget**: Specifies where to display the response after submitting the form,

overriding the form's target attribute. Works with submit and image types.

- formnovalidate**: Specifies that this particular input element should not be validated when submitted,

overriding the form's novalidate attribute. Works with submit type.

Sample form

STUDENT REGISTRATION FORM

First Name

Enter First Name

Last Name

Enter Last Name

Gender

☐ Male ☐ Female

Age

Enter your age

Date of Birth

dd-mm-yyyy

Email Address

Enter your email

Password

Enter your password min 8 characters

Confirm Password

Confirm password

Phone Number

+91

Enter your phone number

Address

Enter your address

State

--Select your state--

Pin Code

Enter your pin code

Hobbies

☐ Music ☐ Movies ☐ Sports ☐ Travel

Upload Photo

*Max size 100kb.

Choose File

No file chosen

☐ I hereby declare that the above information provided is true and correct.

☐ I'm not a robot


reCAPTCHA
Privacy - Terms

Register

Reset

Facebook Registration form

Sign Up
It's free and anyone can join

First Name:

Last Name:

Your Email:

New Password:

I am:

Birthday:

Gmail Registration form



Create your Google Account

First name

Last name

Username

@gmail.com

You can use letters, numbers & periods

[Use my current email address instead](#)

Password

Confirm

Use 8 or more characters with a mix of letters, numbers & symbols

☐

Show password

[Sign in instead](#)

Next

HTML Using Forms

- `<form>` tag
 - Contains the form elements on a web page
 - Container tag
- `<input>` tag
 - Configures a variety of form elements including text boxes, radio buttons, check boxes, and buttons
 - Stand alone tag
- `<textarea>` tag
 - Configures a scrolling text box
 - Container tag
- `<select>` tag
 - Configures a select box (drop down list)
 - Container tag
- `<option>` tag
 - Configures an option in the select box
 - Container tag

Sample Form HTML

<form>

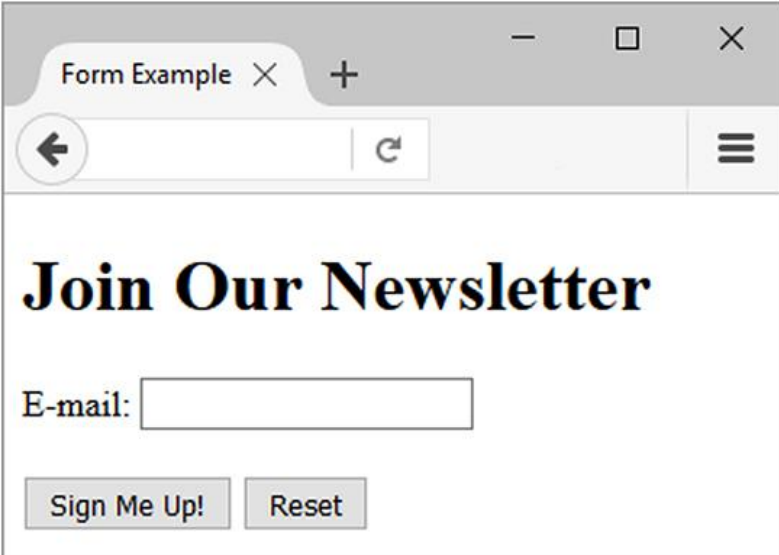
E-mail: <input type="text" name="email" id="email" >

**

**

**<input type="submit" value="Sign Me Up!"> <input
type="reset">**

</form>



The screenshot shows a web browser window with the title 'Form Example'. The browser's address bar is empty. The main content of the page is a form titled 'Join Our Newsletter'. Below the title is the text 'E-mail:' followed by a text input field. At the bottom of the form are two buttons: 'Sign Me Up!' and 'Reset'.

The Form Element <form>

- Container Tag
- The form element attributes:
 - action
 - Specifies the server-side program or script that will process your form data
 - method
 - get – default value, form data passed in URL
 - post – more secure, form data passed in HTTP Entity Body
 - name
 - Identifies the form
 - id
 - Identifies the form

The Input Element Text Box <input>

- Accepts text information
- Common Attributes:
 - type="text"
 - name
 - id
 - size
 - maxlength
 - value
 - required (HTML5)
 - placeholder (HTML5)

Sample Text Box

E-mail:

The Input Element Password Box

<input>

- Accepts text information that needs to be hidden as it is entered
- Common Attributes:
 - type="password"
 - name
 - id
 - size
 - maxlength
 - value
 - required (HTML5)
 - placeholder (HTML5)

Sample Password Box

Password:

The Input Element Check box

<input>

- Allows the user to select one or more of a group of predetermined items
- Common Attributes:
 - type="checkbox"
 - name
 - id
 - checked
 - value

Sample Check Box

Choose the browsers you use:

- ☐ Google Chrome
- ☐ Firefox
- ☐ Microsoft Edge

The Input Element Radio Button

<input>

- Allows the user to select exactly one from a group of predetermined items
- Each radio button in a group is given the same name and a unique value
- Common Attributes:
 - type="radio"
 - name
 - id
 - checked
 - value

Sample Radio Button

Select your favorite browser:

- ☐ Google Chrome
- ☐ Firefox
- ☐ Microsoft Edge

The Textarea Element <textarea>

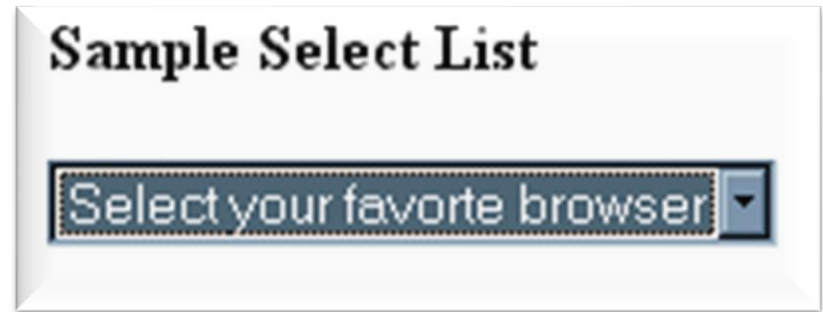
- Configures a scrolling text box
- Common Attributes:
 - name
 - id
 - cols
 - rows

Sample Scrolling Text Box

Comments:

The Select Element <select>

- Configures a select list (along with <option> tags)
- Also known as: Select Box, Drop-Down List, Drop-Down Box, and Option Box.
- Allows the user to select one or more items from a list of predetermined choices.
- Common Attributes:
 - name
 - id
 - size
 - multiple



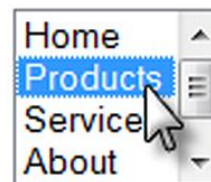
The Option Element <option>

- Configures the options in a Select List
- Attributes:
 - value
 - selected

Select List: One Initial Visible Item



Select List: Four Items Visible



The Input Element Submit Button

<input>

- Submits the form information
- When clicked:
 - Triggers the **action** method on the **<form>** tag
 - Sends the form data (the name=value pair for each form element) to the web server.
- Attributes:
 - type="submit"
 - name
 - id
 - value

Sample Submit Button



The Input Element Reset Button

<input>

- Resets the form fields to their initial values
- Attributes:
 - type="reset"
 - name
 - id
 - value

Sample Reset Button



The Input Element Hidden Field

<input>

- This form control is *not* displayed on the Web page.
- Hidden form fields
 - Can be accessed by both client-side and server-side scripting
- Common Attributes:
 - type="hidden"
 - name
 - id
 - value

The Label Element <label>

- Associates a text label with a form control
- Two Different Formats:

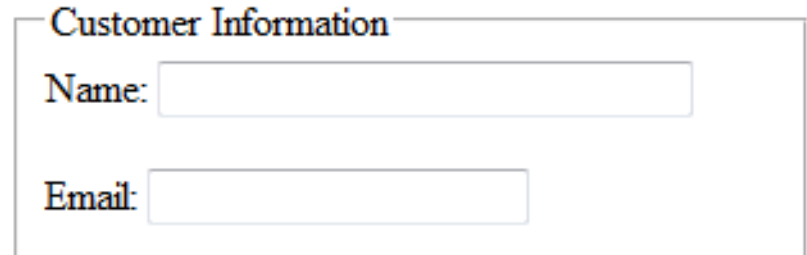
<label>Email: <input type="text" name="CustEmail" id="CustEmail"></label>

OR

**<label for="email">Email: </label>
<input type="text" name="CustEmail" id="email" />**

The Fieldset & Legend Elements

- The Fieldset Element <fieldset>
 - Container tag
 - Creates a visual group of form controls on a web page
- The Legend Element <legend>
 - Container tag
 - Creates a text label within the fieldset



<fieldset><legend>Customer Information</legend>

<label>Name:

<input type="text" name="CName" id="CName" size="30"></label>

**

**

<label>Email:

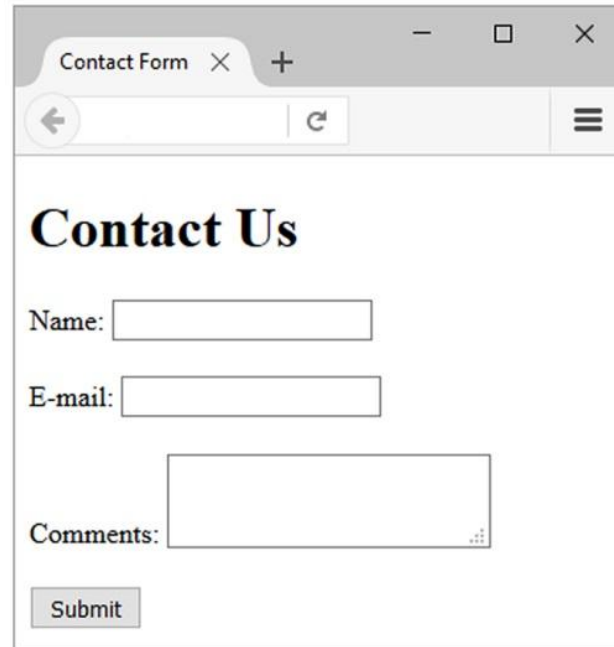
<input type="text" name="CEmail" id="CEmail"></label>

</fieldset>

Using CSS to Style a Form

form

label	text box
label	text box
label	scrolling text box
submit button	



```
form { background-color:#eaeaea; font-family: Arial, sans-serif;  
        width: 350px; padding: 10px;}
```

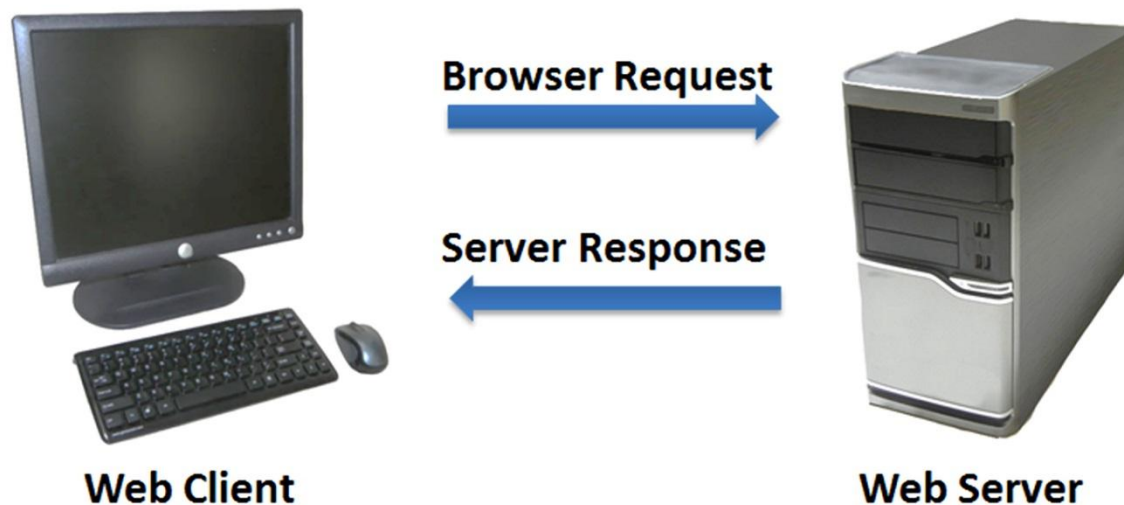
```
label { float: left; clear: left; display: block; width: 100px;  
        text-align: right; padding-right: 10px; margin-top: 10px; }
```

```
input, textarea { margin-top: 10px; display: block;}
```

```
input[type= "submit"]{ margin-left: 110px; }
```

Server-Side Processing

- Your web browser requests web pages and their related files from a web server.
- The web server locates the files and sends them to your web browser.
- The web browser then renders the returned files and displays the requested web pages for you to use.



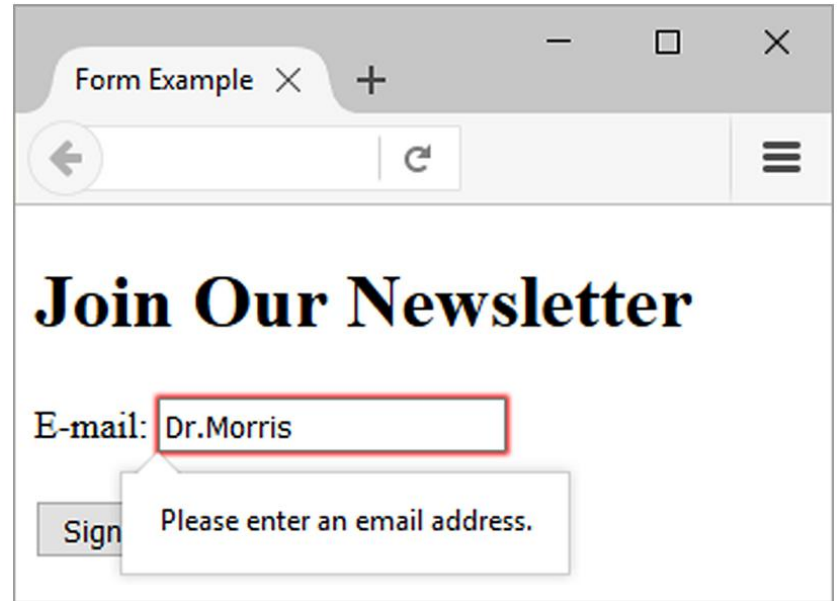
Server-Side Scripting

- One of many technologies in which a server-side script is embedded within a web page document saved with a file extension such as:
 - .php (PHP)
 - .cfm (Adobe ColdFusion)
 - .jsp (Sun JavaServer Pages)
 - .aspx (ASP.Net)
- Uses direct execution — the script is run either by the web server itself or by an extension module to the web server.



HTML5: Email Text Box <input>

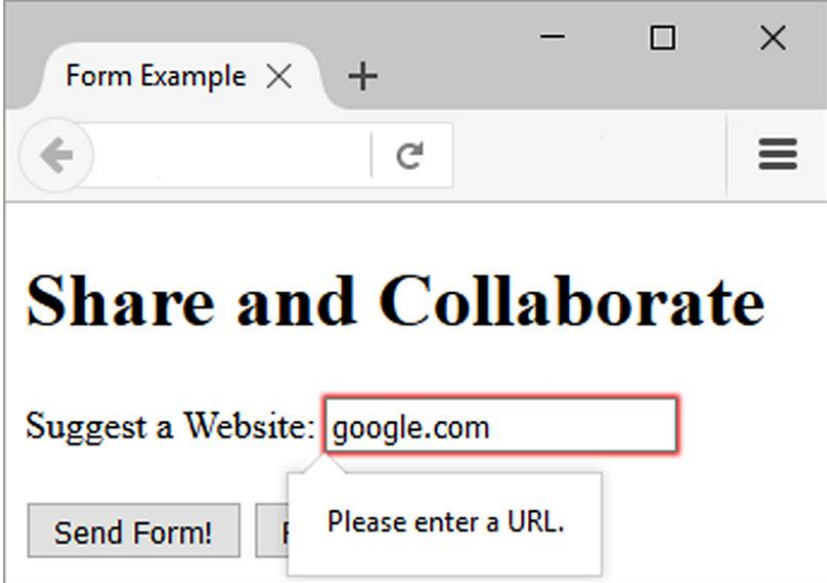
- Accepts text information in e-mail address format
- Common Attributes:
 - type="email"
 - name
 - id
 - size
 - maxlength
 - value
 - placeholder
 - required



The screenshot shows a web browser window with a single tab titled "Form Example". The address bar is empty. The main content of the page is a newsletter sign-up form. At the top, it says "Join Our Newsletter" in a large, bold, black serif font. Below this, there is a label "E-mail:" followed by a text input field. The input field contains the text "Dr.Morris" and is outlined with a red border. Below the input field, there is a "Sign" button. A white tooltip box with a grey border is positioned over the "Sign" button, containing the text "Please enter an email address." in a small, grey font.

HTML5: URL Text Box <input>

- Accepts text information in URL format
- Common Attributes:
 - type="url"
 - name
 - id
 - size
 - maxlength
 - value
 - placeholder
 - required



The screenshot shows a web browser window with a single tab titled 'Form Example'. The address bar is empty. The main content area has the heading 'Share and Collaborate'. Below the heading is a form with the label 'Suggest a Website:'. The text input field contains 'google.com' and is highlighted with a red border. Below the input field is a button labeled 'Send Form!'. A validation message box is displayed below the input field, stating 'Please enter a URL.'.

HTML5: Telephone Number Text Box

<input>

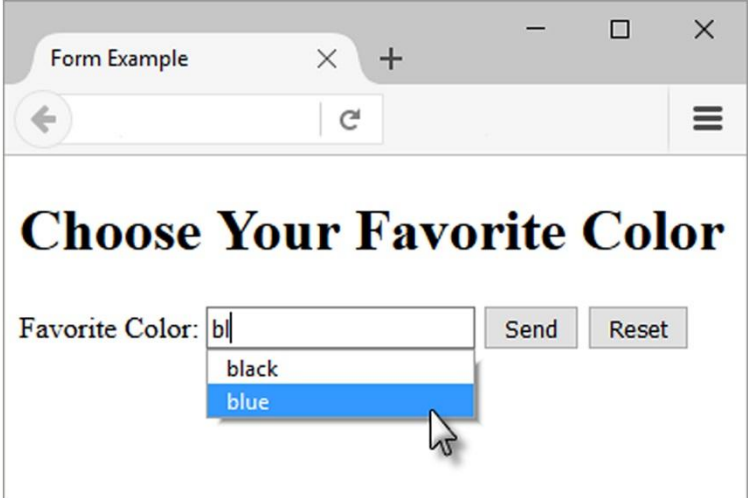
- Accepts text information in telephone number format
- Common Attributes:
 - type="tel"
 - name
 - id
 - size
 - maxlength
 - value
 - placeholder
 - required

HTML5: Search Text Box <input>

- Accepts search terms
- Common Attributes:
 - type="search"
 - name
 - id
 - size
 - maxlength
 - value
 - placeholder
 - required

HTML5: Datalist Control

```
<label for="color">Favorite  
Color:</label>  
<input type="text" name="color"  
id="color" list="colors" >  
<datalist id="colors">  
  <option>red</option>  
  <option>green</option>  
  <option>blue</option>  
  <option>yellow</option>  
  <option>pink</option>  
  <option>black</option>  
</datalist>
```



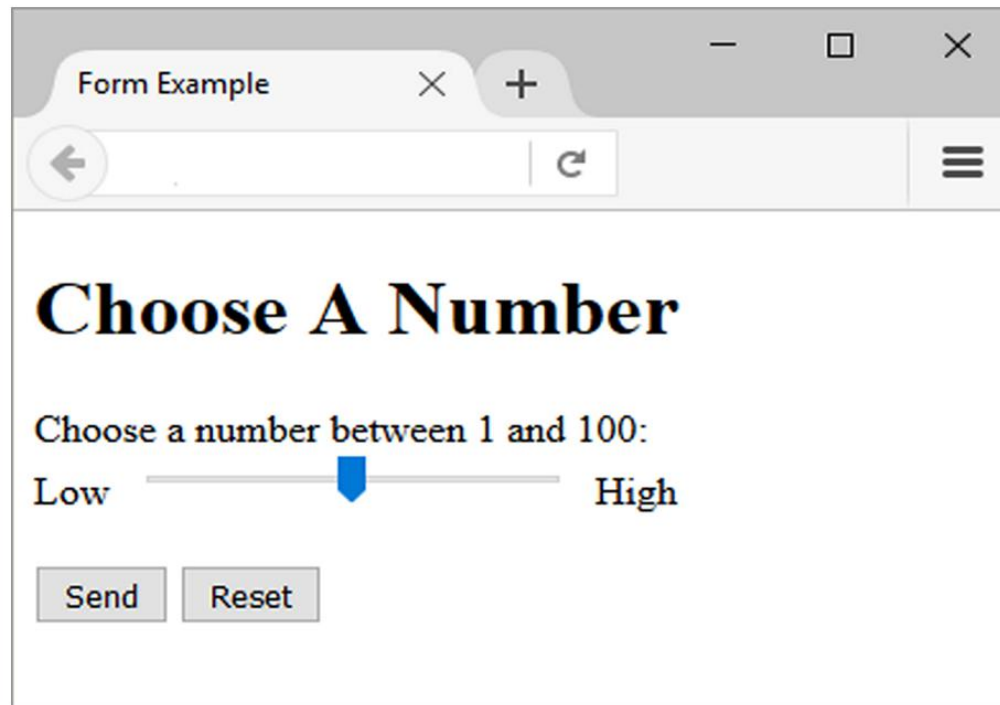
The screenshot shows a web browser window with the title 'Form Example'. The main heading of the form is 'Choose Your Favorite Color'. Below the heading, there is a label 'Favorite Color:' followed by a text input field containing the text 'bl'. A dropdown menu is open below the input field, showing two options: 'black' and 'blue', with 'blue' highlighted in blue. To the right of the input field are two buttons: 'Send' and 'Reset'.

HTML5: Slider Control <input>

<label for="myChoice">

Choose a number between 1 and 100:</label>

Low <input type="range" name="myChoice"
id="myChoice"> High

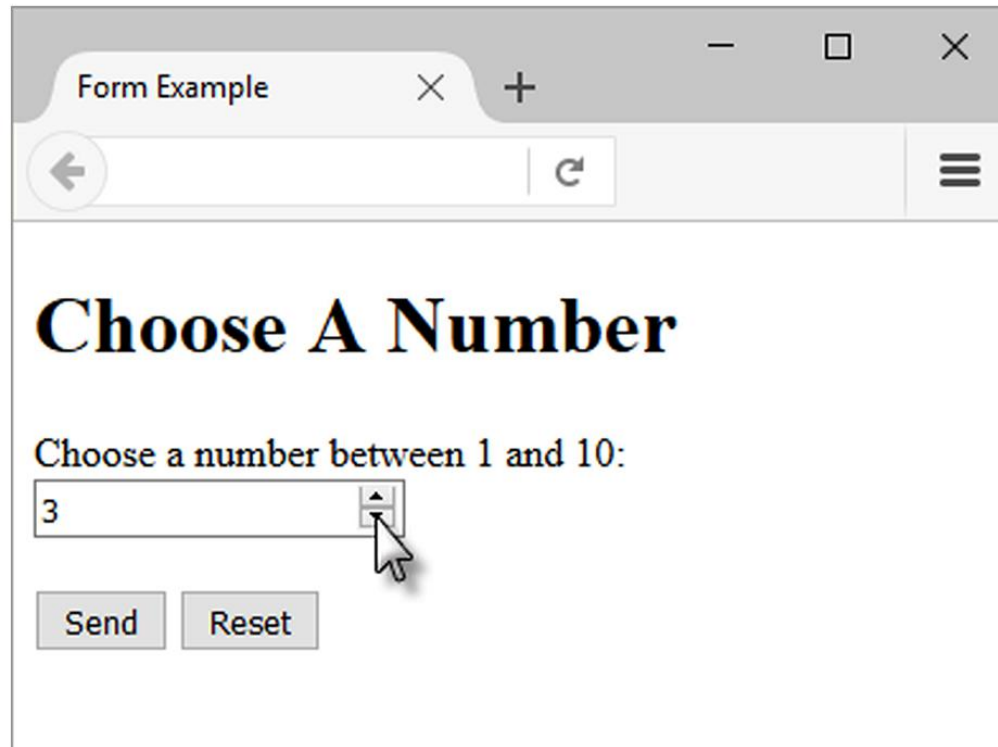
A screenshot of a web browser window titled "Form Example". The browser's address bar is empty. The main content area displays the heading "Choose A Number" in a large, bold, black serif font. Below the heading, the text "Choose a number between 1 and 100:" is shown in a smaller, black serif font. Underneath this text is a horizontal slider control. The slider has a light gray track with a blue triangular handle positioned approximately one-third of the way from the left. The word "Low" is positioned to the left of the track, and the word "High" is positioned to the right. Below the slider, there are two gray rectangular buttons with black text: "Send" on the left and "Reset" on the right.

HTML5: Spinner Control <input>

```
<label for="myChoice">
```

Choose a number between 1 and 10:</label>

```
<input type="number" name="myChoice" id="myChoice"  
      min="1" max="10">
```



The screenshot shows a web browser window with a single tab titled 'Form Example'. The browser's address bar is empty. The main content area of the browser displays a form with the heading 'Choose A Number' in a large, bold, black serif font. Below the heading is a label 'Choose a number between 1 and 10:' in a smaller, black serif font. Underneath the label is a number spinner input field. The field contains the number '3'. To the right of the input field are small up and down arrow buttons for incrementing and decrementing the value. A mouse cursor is positioned over these spinner controls. Below the input field are two rectangular buttons: 'Send' and 'Reset', both with a light gray background and black text.

HTML5: Calendar Control <input>

<label for="myDate">Choose a Date</label>

<input type="date" name="myDate" id="myDate">

Choose a Date

mm/dd/yyyy

Send Reset

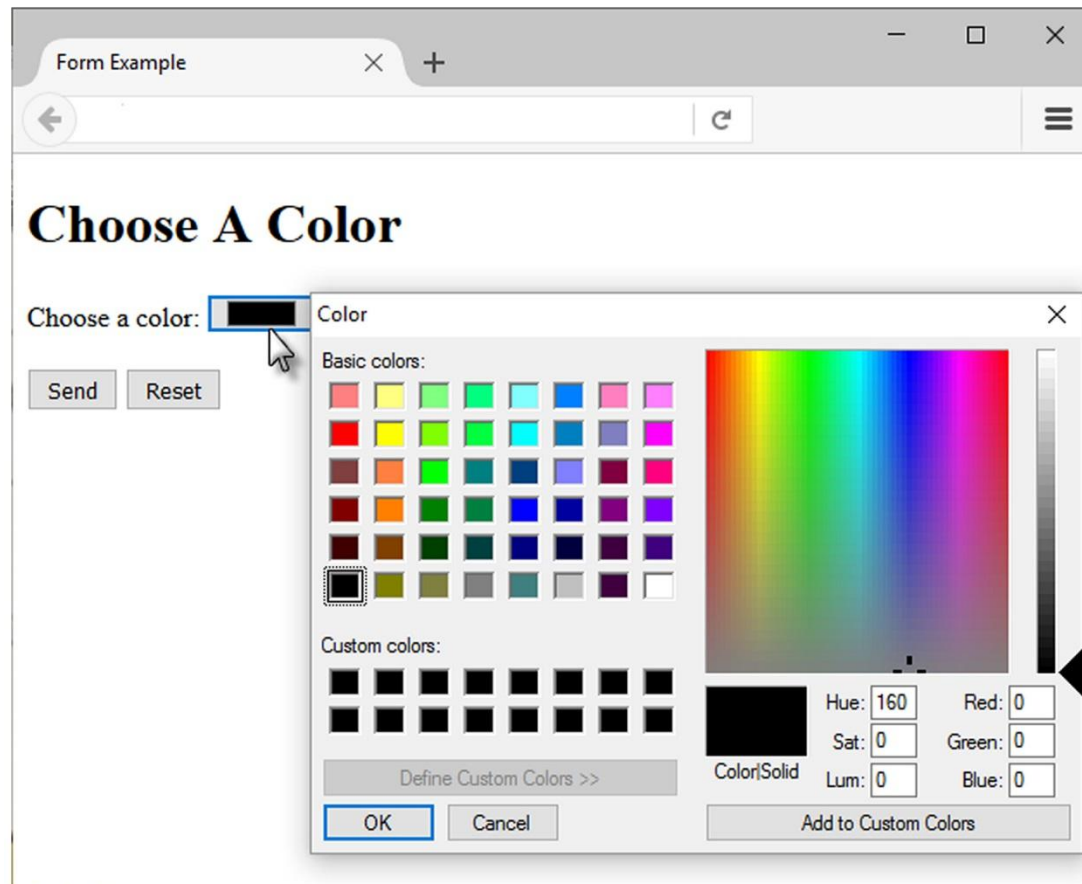
July 2016

Sun	Mon	Tue	Wed	Thu	Fri	Sat
26	27	28	29	30	1	2
3	4	5	6	7	8	9
10	11	12	13	14	15	16
17	18	19	20	21	22	23
24	25	26	27	28	29	30
31	1	2	3	4	5	6

HTML5 Color-well Control

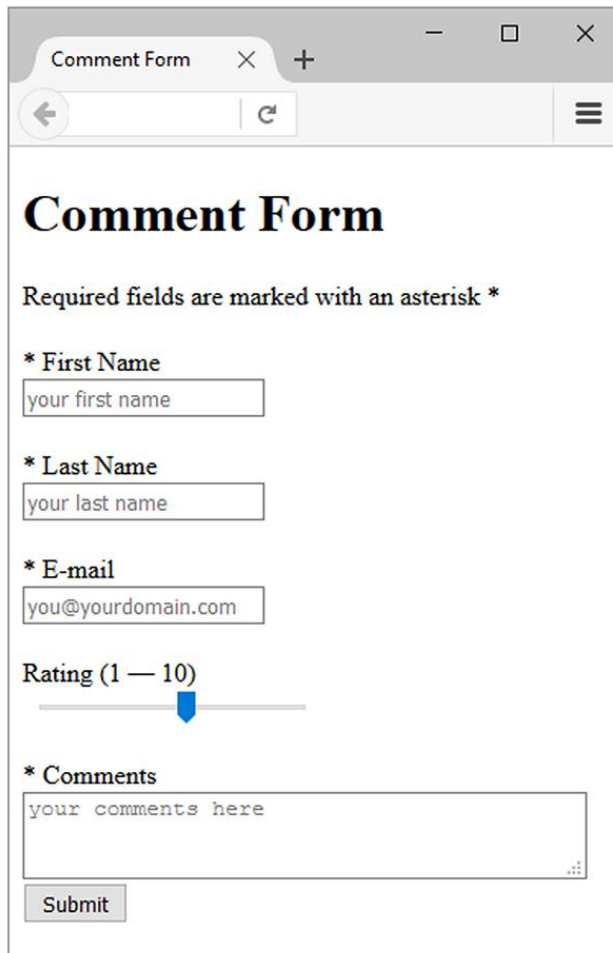
<label for="myColor">Choose a color:</label>

<input type="color" name="myColor" id="myColor">



Practice with an HTML5 Form

- The form display and functioning varies with browser support.



Comment Form

Required fields are marked with an asterisk *

* First Name

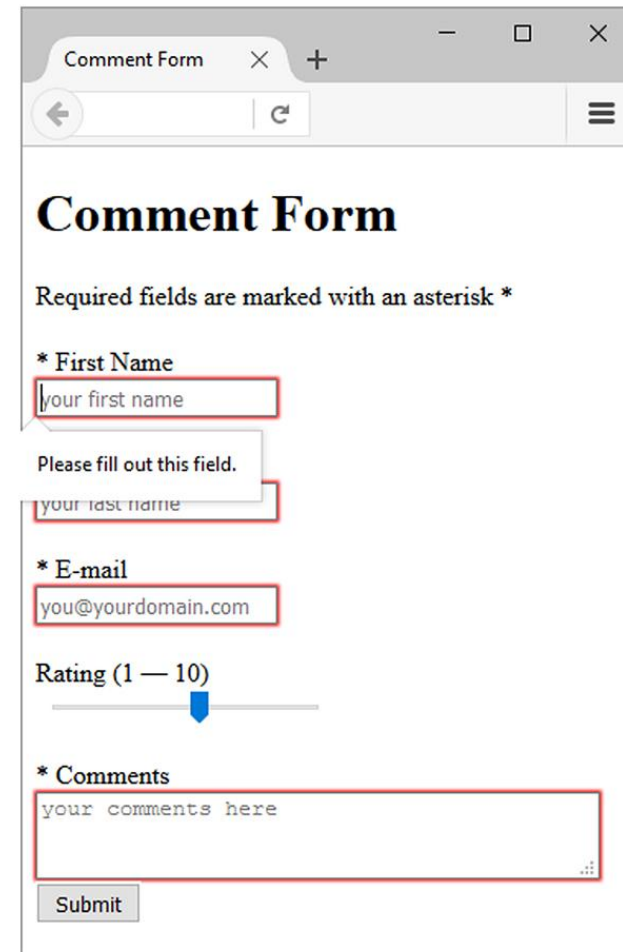
* Last Name

* E-mail

Rating (1 — 10)

* Comments

Submit



Comment Form

Required fields are marked with an asterisk *

* First Name

Please fill out this field.

* E-mail

Rating (1 — 10)

* Comments

Submit

Summary

- This chapter introduced the use of forms on web pages.
- You learned about how to configure form elements and provide for accessibility.
- You learned how to configure a form to access server-side processing.
- You also learned about new HTML5 form controls and attributes.

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